PropertyScope NSW

Release 2 / Technical report

41026 Advanced Software Development — Assessment 3

Submission	Group 20
Team	Matthew Shelton, Burhan Naeem, James Huang, Michael White, Derek Song
Due	25 October 2026, 11:59 pm Sydney time
Repository	MattShelton04/41026ASDProject
Showcase video	YouTube
Commit reference	3078222b79f1c4a53eeaebdf240bc0772d9cba62

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1 Project overview

PropertyScope NSW helps buyers research NSW property. Each student owns a containerised frontend, backend/API and database slice; Matthew also leads the shared platform. Release 2 keeps the Release 0–1 AI-mode, agentic loop, MCP and RAG servers, and adds:

- a host Multi-Agent Server (Planner → Worker → Reviewer → human review), called through a feature's backend;
- three review modes: multi-agent workflow, testing report and cloud deployment;
- pre-commit security scans, and a `student-N.yml` per student for post-commit endpoint tests;
- Azure deployment with the AI tier disabled by default.

Student	Feature	Release 2 responsibility
1 Matthew Shelton (24763373)	F1 Data platform and property discovery	Shared platform (Multi-Agent Server, review modes, security scans, test kit, Azure); F1 workflow and endpoint tests
2 Burhan Naeem (24764134)	F2 Property sales and market cases	Workflow, endpoint tests, evidence
3 James Huang (24970865)	F3 Suburb, crime and liveability analytics	Workflow, endpoint tests, evidence
4 Michael White (24846267)	F4 Site, planning and building due diligence	Workflow, endpoint tests, evidence
5 Derek Song (24833978)	F5 Buyer journey workspace	Workflow, endpoint tests, evidence

2 Final repository structure

Path	Contents	Release
`student-1/` … `student-5/`	Feature slices: UI, API, database service, tools, corpus	0–1
`student-N/config/multi-agent/`, `student-N/tests/endpoints/`	Each feature's workflow template and endpoint tests	2
`ai-services/{ai-mode,agent-core}/`	AI-mode run API and the shared agentic loop	0
`ai-services/{mcp-server,rag-server}/`	Tool transport and local retrieval	1
`ai-services/multi-agent-server/`	Planner, Worker, Reviewer, decisions, JSONL audit, CLI	2
`scripts/devtools/review/`	Review modes and release decision	2
`shared/frontend/multi-agent/`, `shared/testkit/`	Shared workflow panel; endpoint client, JUnit summary, `FakeMultiAgentServer`	2
`.pre-commit-config.yaml`, `scripts/security/`	Security hooks and report	2
`.github/workflows/`	`student-N.yml`, `integration-ci.yml`; `cloud-deployment.yml`	0–2
`deployment/azure/`, `docker-compose.azure.yml`	Bicep, VM scripts, TLS edge, systemd AI units	2
`docs/release-2/evidence/`	Security, CI, multi-agent, review, local and cloud evidence	2

3 Release 2 design flow

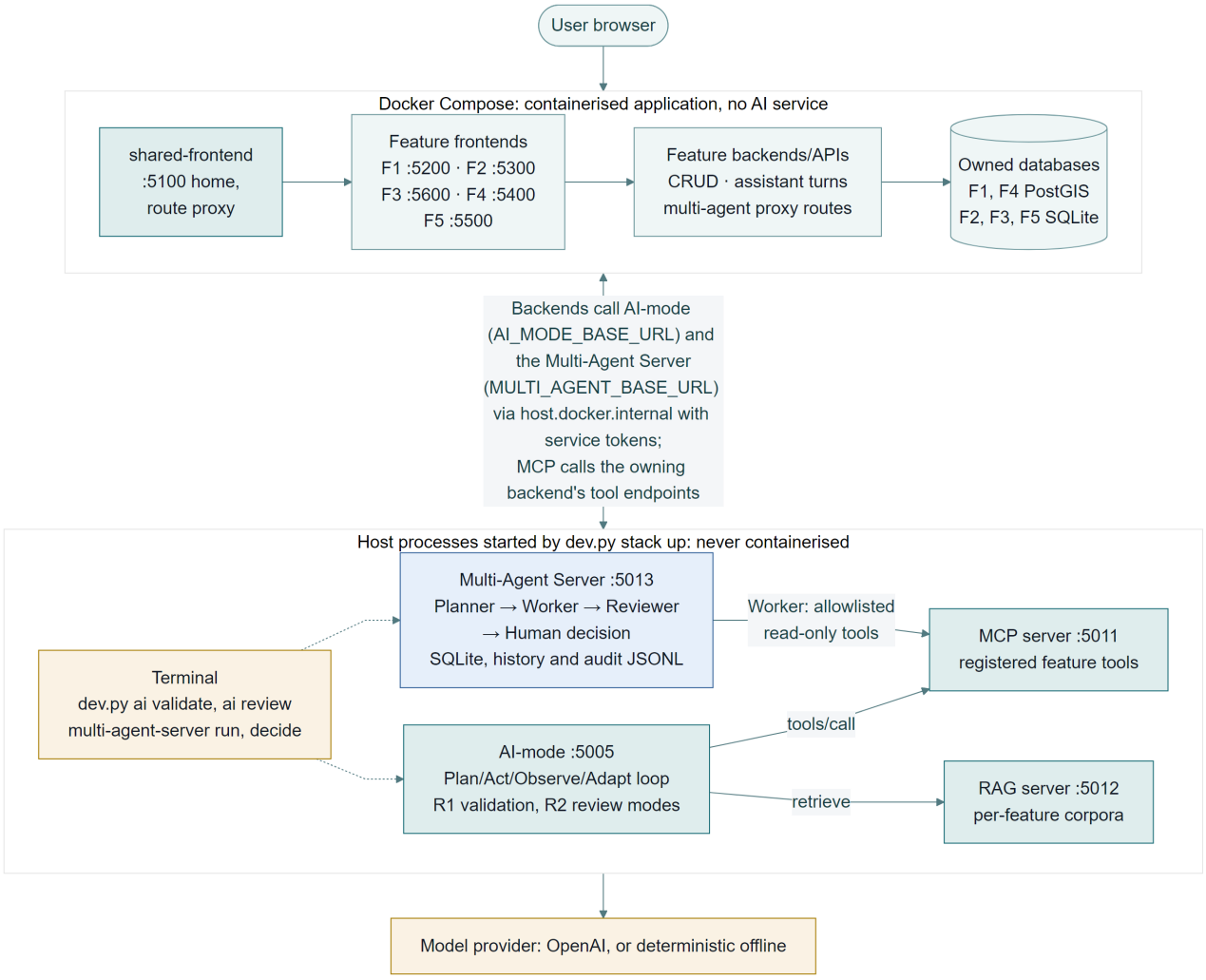


Figure 1 Release 2 integrated architecture

CRUD requests pass from the browser through the shared edge to a feature frontend, its backend/API and its own database service. Assistant turns keep the Release 1 path through AI-mode to the feature's MCP tools and RAG corpus. A backend with a workflow also proxies the Multi-Agent Server over HTTP with a service token through proxy routes that fix the template server-side and hide other features' runs; the shared panel shows each stage up to the human decision. No AI service opens a feature database or runs in Docker.

Student	Frontend → backend/API → database	MCP tools	RAG documents	Multi-agent template (read-only tools)
1	:5200 → `/api/data-platform/v1` → `f1-postgres`	14	19	`f1-release-readiness-review` (2)
2	:5300 → `/api/market-intelligence/v1` → `f2-db-api`	3	3	`f2-market-case-brief` (2)
3	:5600 → `/api/suburb-analytics/v1` → `f3-database`	6	8	`f3-suburb-comparison-review` (2)

Student	Frontend → backend/API → database	MCP tools	RAG documents	Multi-agent template (read-only tools)
4	:5400 → `/api/due-diligence/v1` → `f4-postgres`	3	3	`f4-evidence-readiness-review` (2)
5	:5500 → `/api/buyer-workspaces/v1` → `f5-db-api`	6	5	`f5-buyer-case-action-plan` (1)

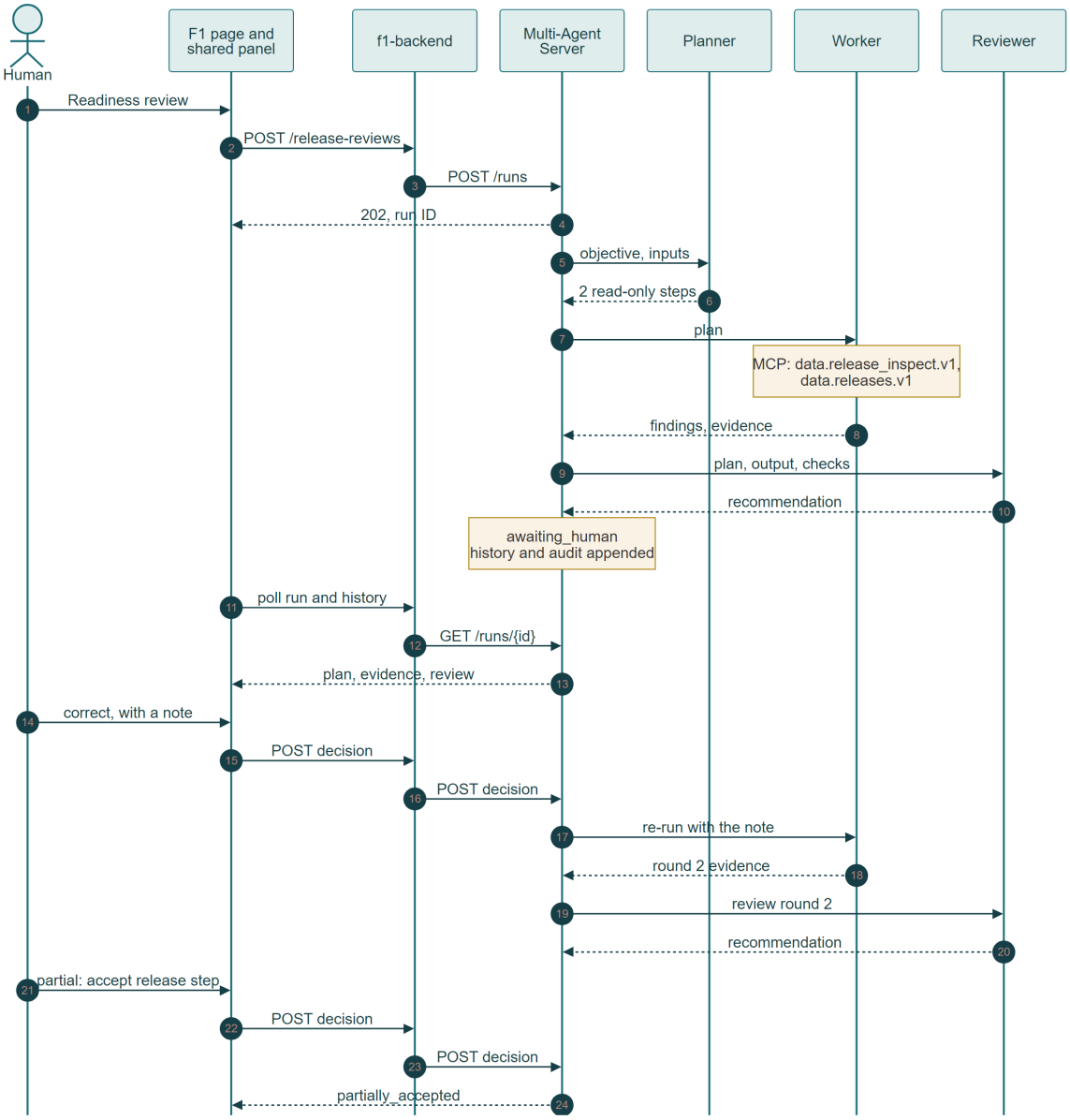


Figure 2 Multi-agent flow, using the Feature 1 readiness review

Student 1

`f1-release-readiness-review` recommends whether a candidate dataset release is ready to publish. Seven proxy routes sit under `/api/data-platform/v1/release-reviews`. The allowlist omits `data.release_publish.v1`, so a review never publishes.

Student 2

`f2-market-case-brief` summarises one saved market case's sale observations. Eight proxy routes sit under `/api/market-intelligence/v1/market-briefs`. Both allowlisted tools, `market.cases.inspect.v1` and `market.sales.summary.v1`, are read-only.

Student 3

`f3-suburb-comparison-review` checks one saved suburb comparison against its published crime evidence. Eight proxy routes sit under `/api/suburb-analytics/v1/comparison-reviews`. Both allowlisted tools, `suburb.comparison.inspect.v1` and `crime.saved-comparison-evidence.v1`, are read-only; a review never ranks suburb safety.

Student 4

`f4-evidence-readiness-review` drafts a professional's verification questions from one saved site review's evidence. Eight proxy routes sit under `/api/due-diligence/v1/evidence-reviews`. Both allowlisted tools, `duediligence.review.inspect.v1` and `duediligence.evidence.summary.v1`, are read-only.

Student 5

`f5-buyer-case-action-plan` proposes follow-ups from one buyer case's shortlist, notes, tasks and research. Eight proxy routes under `/api/buyer-workspaces/v1/buyer-cases/{id}/evidence-reviews` recheck case ownership. Its one read-only tool, `buyer.action-plan.context.v1`, takes only a section; the backend, not the model, binds the case from the run.

4 Cloud deployment and CI/CD architecture

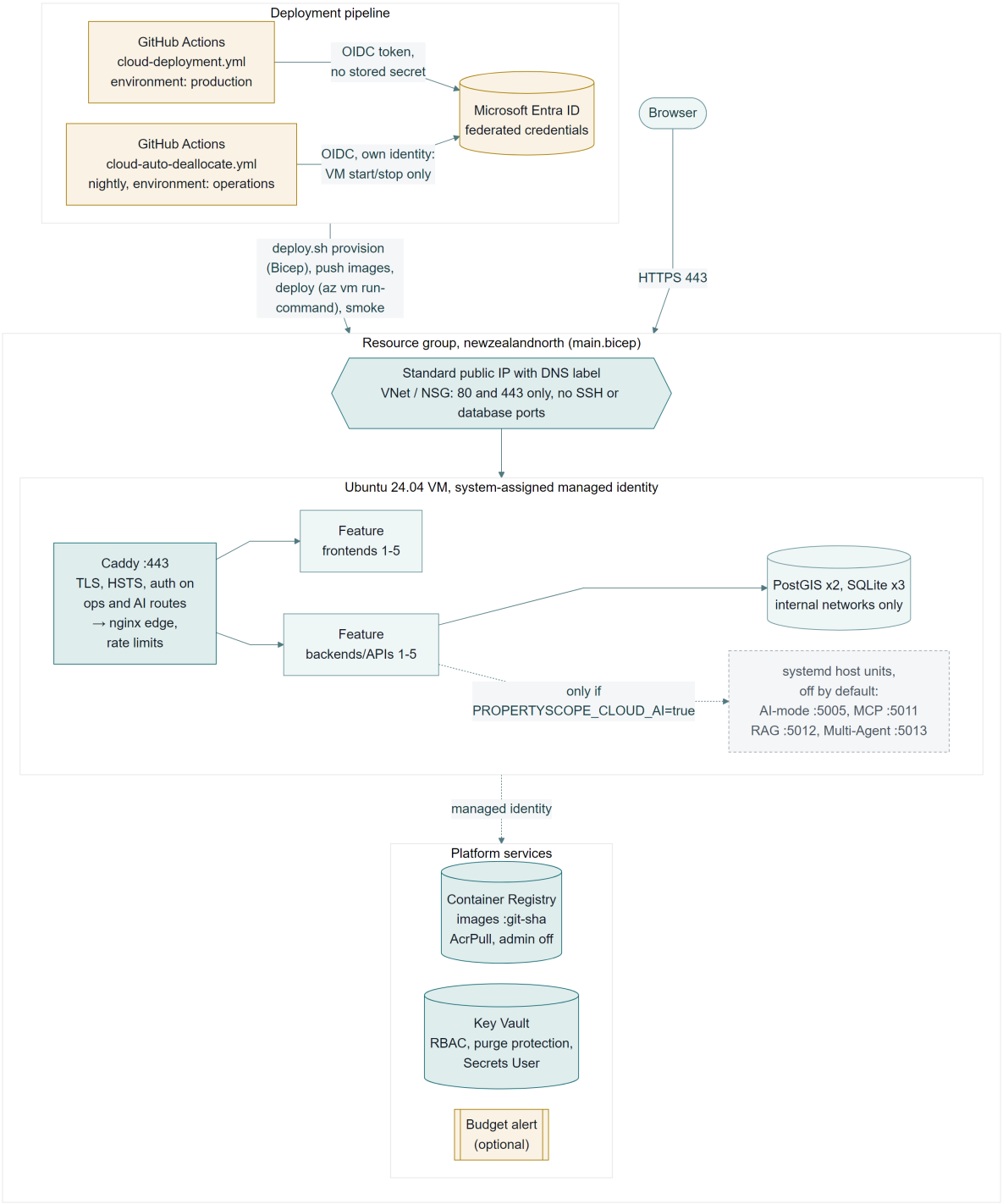


Figure 3 Azure reference architecture

We chose Azure (ADR-048): one Ubuntu VM (`Standard_D4as_v4`, `newzealandnorth`) runs the Compose model with a production override, because the AI tier must stay a host process. Its managed identity reads Container Registry and Key Vault; GitHub signs in with OIDC. The NSG admits only 80 and 443, Caddy terminates TLS, and the four AI systemd units stay off unless `PROPERTYSCOPE_CLOUD_AI=true`.

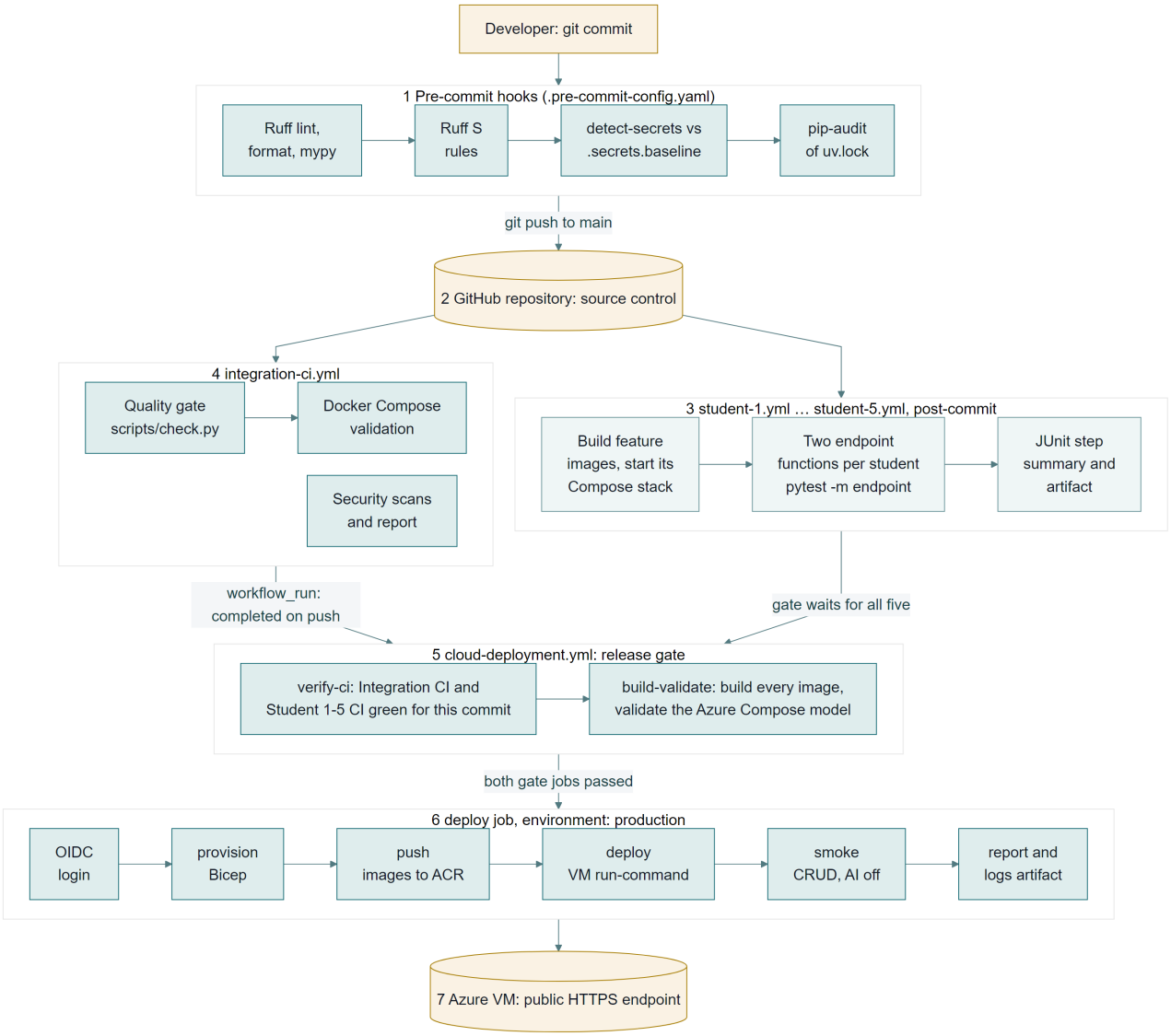


Figure 4 DevOps flow from commit to Azure

A commit runs the pre-commit hooks; a push to `main` runs `integration-ci.yml` and the five `student-N.yml` workflows, whose endpoint tests are each owner's deliverable. `cloud-deployment.yml` gates the release: `verify-ci` waits until all six pass for the commit and `build-validate` builds and validates every image. Only then does `deploy` run these reusable scripts in the protected `production` environment:

Script or template	Role
`deploy.sh provision`, `secrets`	Resource group, `main.bicep`, missing Key Vault secrets
`deploy.sh push`, `deploy`	Images to ACR; VM converged through `az vm run-command`
`deploy.sh smoke`, `status`, `logs`	Public smoke test and VM state
`deploy.sh ai on/off`, `ai-probe`	Bonus AI tier and its B1–B4 probe
`deploy.sh validate-endpoint`, `validate-data`	B5 and B6 checks
`main.bicep`, `modules/`	Network, registry, Key Vault, VM, RBAC, shutdown, budget
`validate-iac.sh`	Offline Bicep, shell, Caddy and nginx checks (CI `iac-lint`)

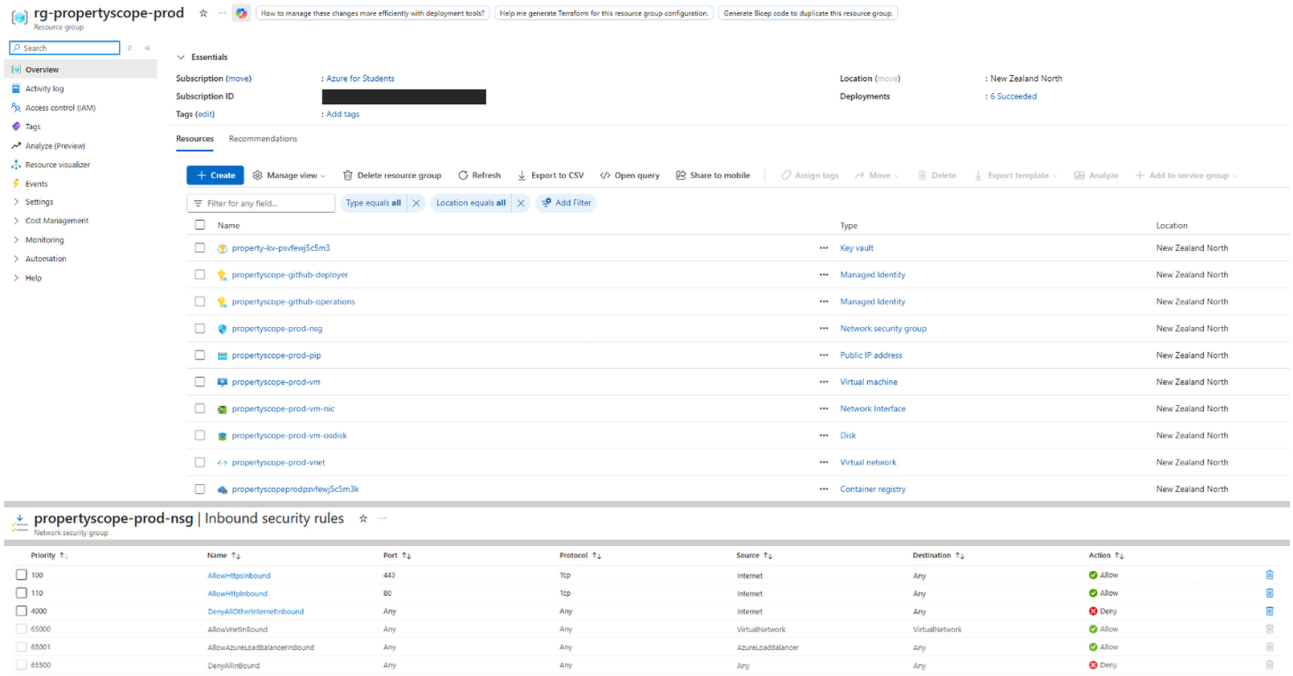


Figure 5 Deployed resources and inbound rules

5 Pre-commit security testing

`.pre-commit-config.yaml` runs three security hooks on every `git commit`: Ruff S (flake8-bandit) rules, detect-secrets against the reviewed `.secrets.baseline`, and pip-audit of `uv.lock`. `integration-ci.yml` repeats them on every push. `dev.py security report` records results per slice and owner:

Security report: overall **PASS**, commit `7b3fb43`, 2026-10-23.

Scan	Result	Findings
Ruff	**PASS**	0 open; 27 fixed; 206 justified with `# noqa`; 0 pending owner
detect-secrets	**PASS**	0 new; 0 confirmed; 0 unaudited; 57 reviewed false positives in `.secrets.baseline`
pip-audit	**PASS**	128 locked packages audited; 0 open; 55 accepted risk(s) in 2 package(s)

Slice	Owner	Ruff open	Fixed	Noqa	Pending owner	Secrets to act on
Shared platform	Matthew Shelton	0	18	145	0	0
Student 1	Matthew Shelton	0	9	26	0	0
Student 2	Burhan Naeem	0	0	2	0	0
Student 3	James Huang	0	0	10	0	0
Student 4	Michael White	0	0	6	0	0
Student 5	Derek Song	0	0	17	0	0

In the recorded commit run, detect-secrets first blocked the commit; after seven false positives were reviewed into the baseline, it passed:

```
security: Ruff flake8-bandit rules (S).....................Passed
security: detect-secrets against .secrets.baseline.............Passed
security: pip-audit of locked dependencies (uv.lock)............Passed
```

6 Post-commit CI/CD testing

Each owner's `student-N.yml` starts the feature's Compose stack with AI disabled, runs `pytest -m endpoint` on two endpoint functions and publishes the Shared test kit's JUnit summary (at least two passes). `collect_ci_evidence.py` keeps each run's record, reports and log.

Student	Successful `student-N.yml` run	Endpoint tests
1	#415	46 passed, 0 failed (2 reports) (record, log)
2	#198	4 passed, 0 failed (record, log)
3	#192	4 passed, 0 failed (record, log)
4	#186	10 passed, 0 failed (record, log)
5	#195	4 passed, 0 failed (record, log)

Every run is a push to `main` at `7b3fb43`.

Student 1

`student-1.yml` loads fixture data, then tests the feature origin (:5200) and the shared edge (:5100):

1. `GET /api/data-platform/v1/properties/search`: contract-valid ranked matches for an accepted address; invalid queries return structured problems.
2. `/api/data-platform/v1/sources`: create, list, read, update and delete a source definition; invalid, duplicate and non-JSON bodies are rejected and not persisted.

Student 2

`student-2.yml` tests two endpoint functions on the feature origin (:5300):

1. `POST /api/market-intelligence/v1/market-cases`: a valid case is persisted at version 1 and readable at its own URL; a window that ends before it starts is refused `422 invalid_market_case` and writes nothing.
2. `PUT /api/market-intelligence/v1/market-cases/{id}`: the current version applies the change and advances the version; a stale version is refused `409 version_conflict` and leaves the first writer's value in place.

Student 3

`student-3.yml` tests two endpoint functions on the feature origin (:5600):

1. `GET /api/suburb-analytics/v1/suburbs`: `limit=2` returns a bounded page of two NSW suburbs from at least ten seeded; a non-numeric limit returns a `422` Problem Details response.

2. `POST /api/suburb-analytics/v1/suburb-comparisons`: a valid comparison is persisted at version 1, readable at its own URL, then deleted (`404 comparison_not_found`); an unknown measure is refused `422 invalid_request` and writes nothing.

Student 4

`student-4.yml` tests two endpoint functions on the feature origin (:5400):

1. `POST /api/due-diligence/v1/site-reviews`: a valid review is persisted as a version 1 draft, readable at its own URL and then deleted; five invalid bodies and a non-JSON body are refused `422 invalid_site_review` and write nothing.
2. `GET /api/due-diligence/v1/site-reviews/{id}/evidence`: a seeded review returns schema-valid constraint and building records for its own property only; an unknown or deleted review returns `404 site_review_not_found`.

Student 5

`student-5.yml` tests two endpoint functions on the feature origin (:5500):

1. `POST /api/buyer-workspaces/v1/buyer-cases`: a valid case is persisted active at version 1, readable at its own URL, echoes the request ID and hides `owner_ref`; an inverted budget is refused `422 validation_failed`.
2. `POST /api/buyer-workspaces/v1/buyer-cases/{id}/tasks`: a task is persisted incomplete at version 1, readable and listed under its case; an unknown case returns `404 resource_not_found`.

7 Testing review

`dev.py ai review testing` checks the security report and each `ci/student-N.md` against a deterministic checklist and through the shared loop; the stricter verdict wins.

Verdict	Checklist	Engine	Loop run	Date	Validation log
pass_with_risks	36/37 passed	AI-mode, openai / gpt-5.6-terra	`9fe0c52d`	2026-10-24	5 entries

Latest validation-log entry: `review_completed`, review `1841531b`, `review-testing.v2`, bundle `a669ffbc`.

> Complete owners are student-1, student-2, student-3, student-4 and student-5; all submitted green CI endpoint evidence with run links, commit SHAs and no recorded JUnit failures or errors. (full review)

Finding (medium, shared, `security.dependencies-clean`): The shared dependency audit records 73 known vulnerabilities, including multiple Pillow 11.3.0 advisories.

8 Local deployment evidence

`dev.py stack up` starts the four host AI services, then the Compose application; backends reach them through `host.docker.internal`. `capture_local_deployment.py` checks the running system, including that each AI service is a host process in its own terminal, not a container.

Local deployment report: **PASSED**, 111 of 111 checks passed, captured 2026-10-23 at commit `7b3fb43`.

Check group	Passed
Compose containers	22
Host AI services	9
Edge, frontends and health	13
CRUD round trips	22
Data	1
Direct AI probes	32
Multi-Agent terminal run	6
Evidence files	6

Terminal	Command	Log
Docker Compose stack	`uv run scripts/dev.py stack up; stack logs`	terminal-1-compose.log
AI-mode host process	`uv run scripts/dev.py ai serve ai-mode`	terminal-2-ai-mode.log
MCP host process	`uv run scripts/dev.py ai serve mcp`	terminal-3-mcp.log
RAG host process	`uv run scripts/dev.py ai serve rag`	terminal-4-rag.log
Multi-Agent host process	`uv run scripts/dev.py ai serve multi-agent`	terminal-5-multi-agent.log
System test	`uv run python scripts/capture_local_deployment.py`	terminal-6-system-test.log

9 Cloud deployment report and review

`deploy.sh smoke` checks the home page, every feature route, create, read and delete per feature, and that AI-mode answers `503 ai_disabled`; all 37 checks passed. `dev.py ai review cloud` reviews the evidence, then a named person records the release decision with `ai review decide`, which refuses if the evidence changed.

Item	Recorded
Deployment	**SUCCEEDED** (report)
Public URL	https://propertyscope-nsw-g20.newzealandnorth.cloudapp.azure.com
Commit / image tag	`7b3fb43728582d21bb8c9722405c040e3c18cbf2` / `7b3fb43728582d21bb8c9722405c040e3c18cbf2`
Workflow run	https://github.com/MattShelton04/41026ASDProject/actions/runs/40041985429
AI tier	flag off (PROPERTYSCOPE_CLOUD_AI), edge reports disabled

Item	Recorded
Human release decision	**RELEASE** by Matthew Shelton, 2026-10-24 (record)

Run 40041985429 deployed the submitted code: `verify-ci` 4 min 37 s (waiting for this commit's CI), `build-validate` 45 s, `deploy` 9 min 28 s (run record, step logs, `vm-deploy.log`). Run 39098352903 also ran the Bicep provision step (1 min 29 s).

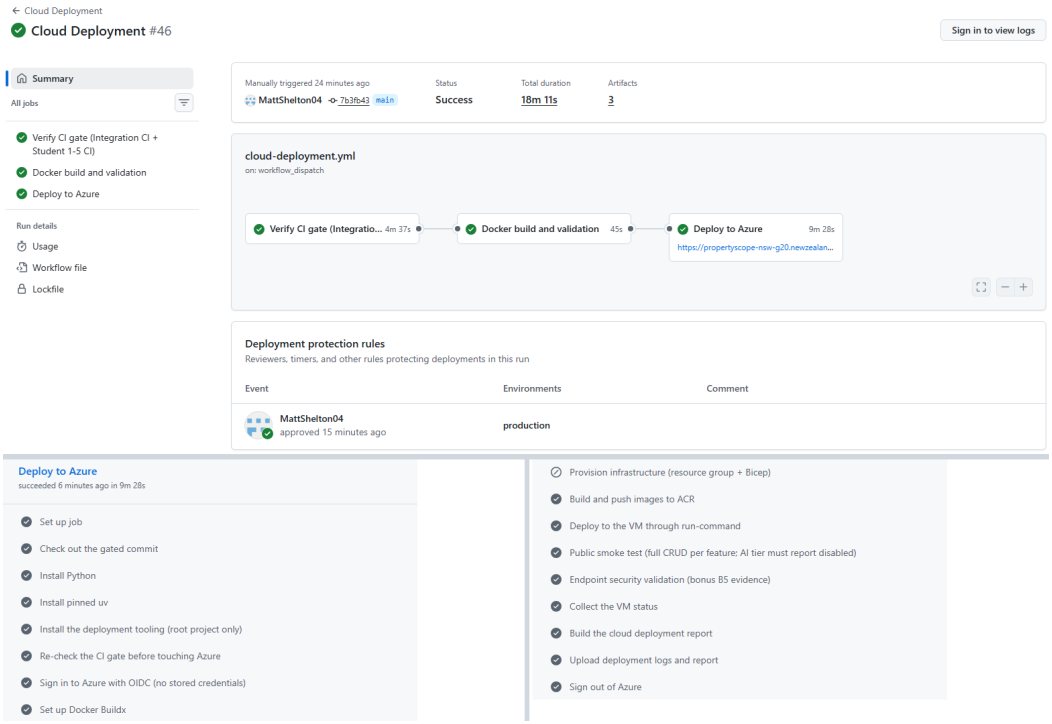


Figure 6 Gated cloud deployment run

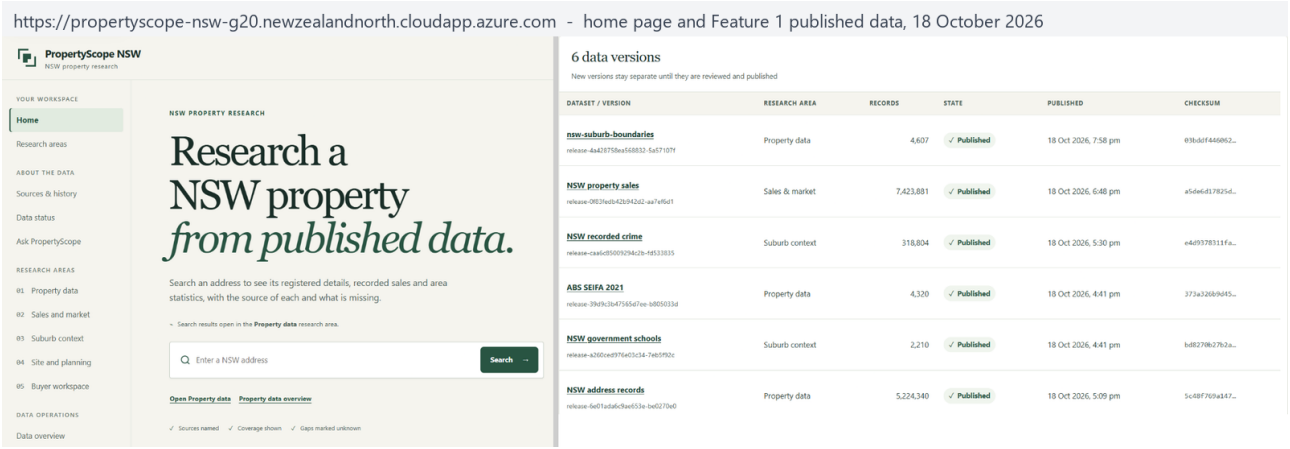


Figure 7 Deployed application with real Feature 1 data

Verdict	Checklist	Engine	Loop run	Date	Validation log
pass	18/18 passed	AI-mode, openai / gpt-5.6-terra	`ce7e9072`	2026-10-24	5 entries

Latest validation-log entry: `human_release_decision` (**release**), review `be5514a8`, `review-cloud.v2`, bundle `ebc3e195`.

> All 18 checklist checks passed. (full review)

10 Multi-agent workflow evidence

The server keeps runs in SQLite, state transitions in `workflow_history.jsonl`, and handoffs, model and tool calls and decisions, with actor and time, in `coordination_audit.jsonl`; `multi-agent-server export` copies a run into the evidence.

Student	Run	History events	Audit events	Handoffs	Tool calls	Decisions	Final state
1	18740ca5	8	24	7	4	correct → partial	partially_accepted
2	f1c290a2	8	29	7	4	correct → approve	approved
3	5338d71c	8	29	7	4	correct → reject	rejected
4	2956e465	8	32	7	4	correct → approve	approved
5	e9600341	8	33	7	8	correct → approve	approved

Other exported runs in the evidence folders: 7.

Multi-agent workflow review

`dev.py ai review multi-agent` checks every export for all four stages, legal transitions, attributed human decisions and allowlisted tool calls.

Verdict	Checklist	Engine	Loop run	Date	Validation log
pass	35/35 passed	AI-mode, openai / gpt-5.6-terra	c12d1292	2026-10-24	7 entries

Latest validation-log entry: `review_completed`, review `47928330`, `review-multi-agent.v2`, bundle `11815455`.

> Students 1, 2, 3, 4 and 5 are complete; no owner evidence is missing. (full review)

Student 1

UI run `18740ca5` (10 October, model provider) reviewed a real 680-record ABS CPI candidate. Round 1: the Reviewer recommended approve; the human chose correct, asking for completeness against ABS. Round 2: the Reviewer recommended correct, as read-only tools cannot fetch ABS totals; the human accepted step `release` only, publishing nothing. Terminal run `c17c66cc` used the same template:

```
$ multi-agent-server export c17c66cc-6b40-4685-909a-440401ae8ec7
09:41:41 r1              - -> planning           multi-agent-server
09:41:52 r1       planning -> working            planner
09:42:03 r1        working -> reviewing          worker
09:42:07 r1      reviewing -> awaiting_human     reviewer
09:42:26 r2 awaiting_human -> working            matthew
09:42:42 r2        working -> reviewing          worker
09:42:49 r2      reviewing -> awaiting_human     reviewer
09:43:11 r2 awaiting_human -> partially_accepted matthew
```

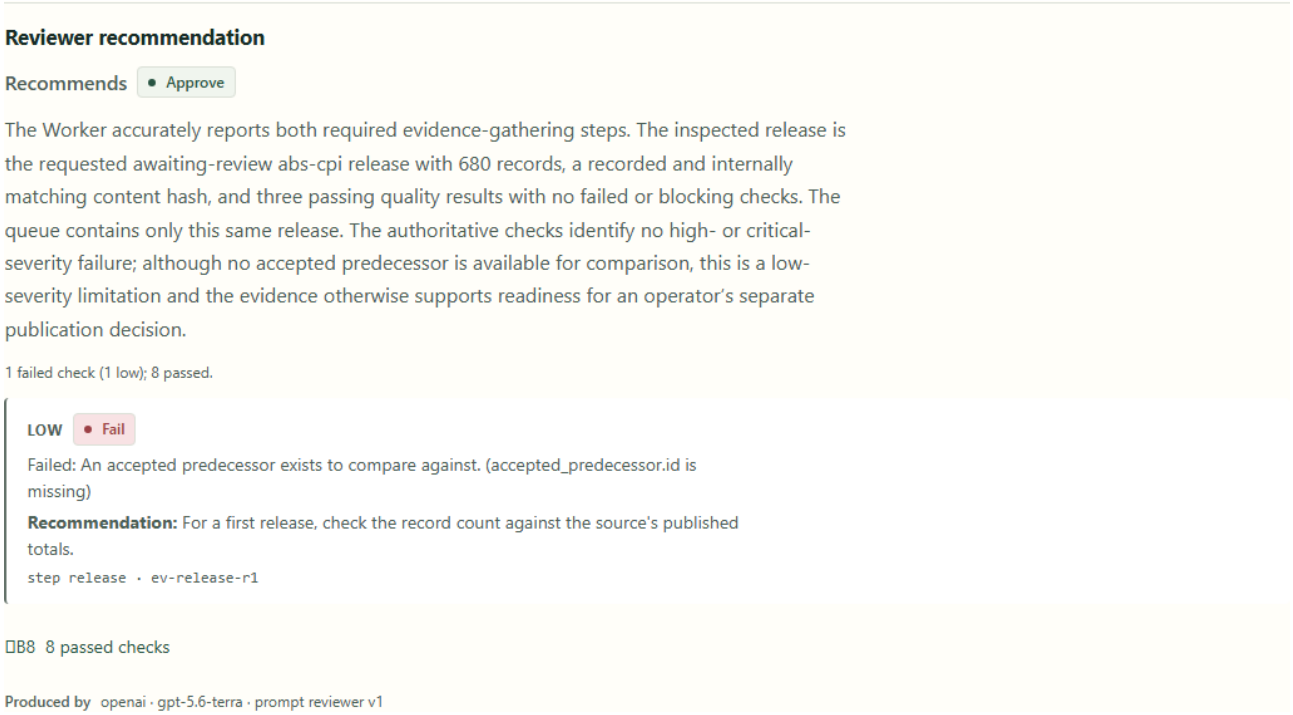


Figure 8 F1 Reviewer recommendation, round 1

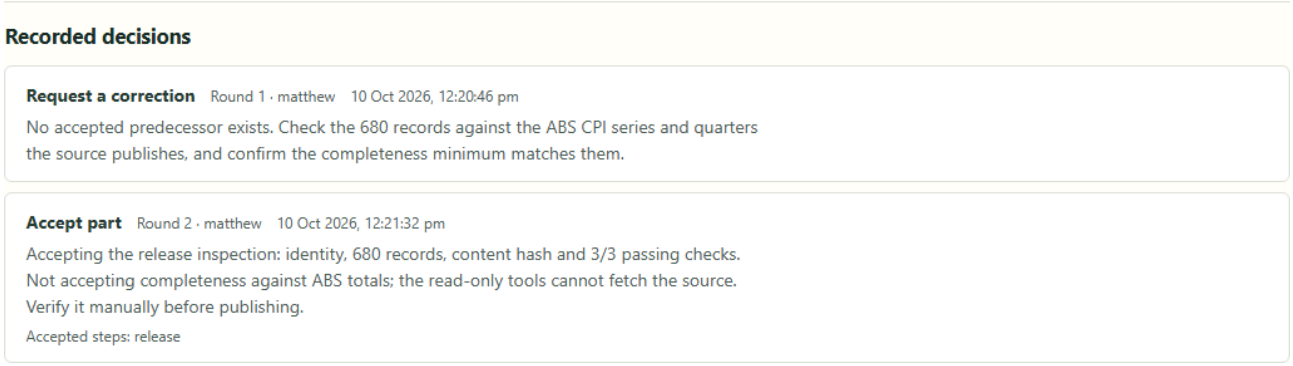


Figure 9 F1 recorded human decisions

Student 2

UI run `f1c290a2` briefed `Sydney example - full history`. The Reviewer recommended approve on all eleven checks passing. Burhan sent it back: the checks read the tool results, while the prose summary reported the figures without the limitations. Round two restated them with the synthetic-data disclosure, and Burhan approved it, 0 failed and 11 passed.

Student 3

UI run `5338d71c` reviewed seeded comparison `comparison-10` (Newcastle and Wollongong, rate). The Reviewer recommended correct: coverage was unavailable, the denominator invalid and the seeded-data disclosure missing. It was sent back; round two added the disclosure, the Reviewer again recommended correct, and it was rejected until counts are used or the missing BOCSAR months are published.

Student 4

UI run 2956e465 reviewed seeded review 1 (11 Example Street, Sydney). The Reviewer recommended approve on all ten checks passing. Michael sent it back: the draft referred to the saved question list, missed two conflicting records and did not plainly disclose seeded data. Round two fixed all three, and Michael approved it, noting three questions still to write.

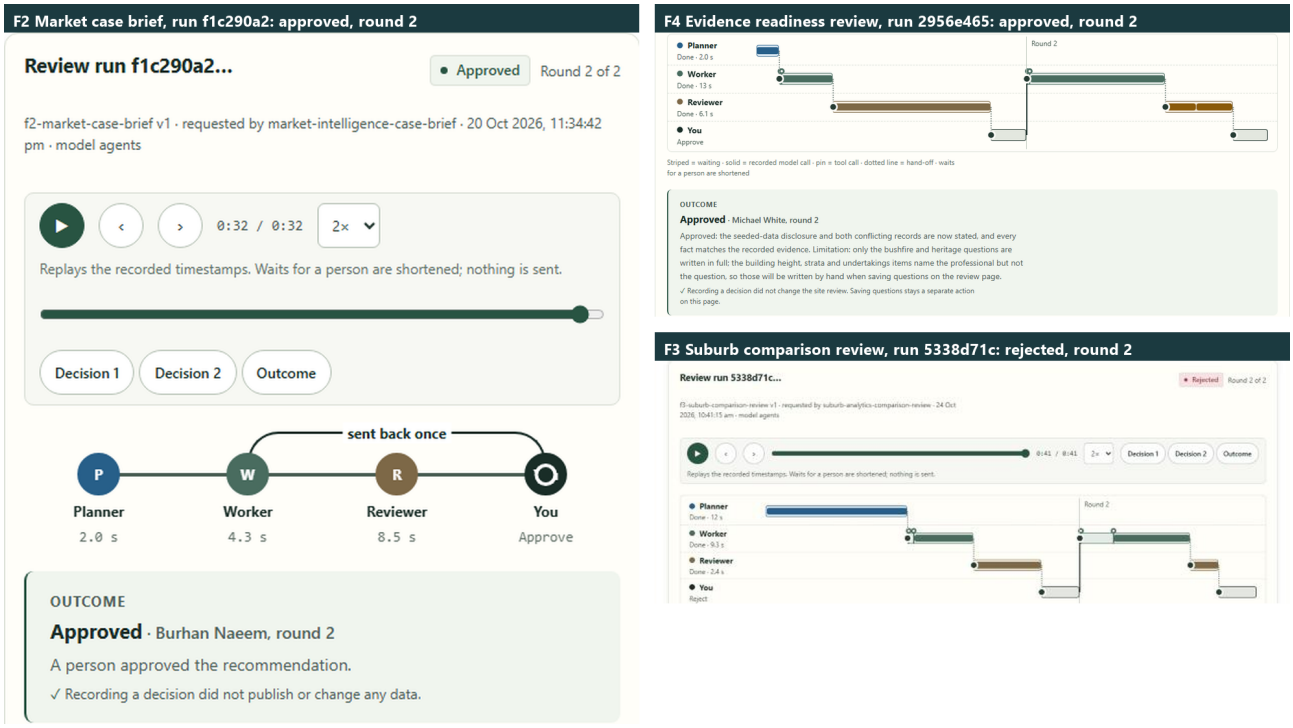


Figure 10 F2–F4 workflow panels in the live app

Student 5

UI run e9600341 reviewed seeded case `Wollongong coastal review`. The Reviewer recommended correct: the Worker proposed no actions. Derek sent it back; round two still had none, and the Reviewer recommended partial (one high, two medium failures). Derek approved it anyway.

11 GitHub commit logs

Commit history: https://github.com/MattShelton04/41026ASDProject/commits/main. Section 12 lists each student's commits.

12 Contribution logs

Student 1 Matthew Shelton

Date	Commit	Release 2 work
8–10 Oct	36a4c26, f86f787, bef57da, 06e0ed0	Multi-Agent Server, review modes, security scans, test kit, panel, F1 review
15–17 Oct	d449a43, 07b1c81, 72f3272, fdc5111, 93d0f53	CI and local evidence, report builder, release gate
17 Oct	5e90231, ad2b443, 71764a5, cd8e56d	Azure switch-on, site login, security fixes
18 Oct	b638c86, 53a2143, 65de8f6, b71c49c, 3c25c08, c1dab31, 3ea2932, 817ef77, 0cf5e3f	F1 suburb-map API, real data and Cloud AI Tier on Azure, cloud and bonus evidence, release decision
19 Oct	daee27d, 70f3b9a	Report against the rubric, B1–B4 UI captures
22–24 Oct	002691e, d7602b7, 81c833d, 7b3fb43	CVE fixes, review bundle limits, CI resilience, cold-start smoke retry

Student 2 Burhan Naeem

Date	Commit	Release 2 work
21 Oct	4162bbd	Market case brief: workflow template, eight proxy routes, panel; endpoint tests in `student-2.yml`; Ruff S findings cleared
21 Oct	24e6025	CI evidence and Feature 2 report sections

Student 3 James Huang

Date	Commit	Release 2 work
22 Oct	08a9331	Suburb comparison review: workflow template, two read-only tools, eight proxy routes, panel; endpoint tests in `student-3.yml`; edge protection and Azure Multi-Agent settings
23 Oct	bc5e46e	Suburb boundary and amenity map from Feature 1, saving from the comparison page

Student 4 Michael White

Date	Commit	Release 2 work
21 Oct	58def77	Endpoint tests in `student-4.yml`; Ruff S findings cleared
21 Oct	aa9b866	Evidence readiness review: workflow template, eight proxy routes, review-page panel; Release 1 question-extraction fix
21 Oct	37b2ffc	CI, security and multi-agent evidence; Feature 4 report sections

Student 5 Derek Song

Date	Commit	Release 2 work
23 Oct	a400cb0	Buyer case action plan: workflow template (v1, v2), case-bound read-only tool, eight proxy routes, panel; endpoint tests in `student-5.yml`; Ruff S findings cleared; security and multi-agent evidence

13 Bonus evidence

We claim B1–B6. B1–B4 ran through the public edge in Cloud AI Tier run 39119158736 (`817ef779`, 18 October), and again before the showcase in 39720293518 (`dedda460`, 22 October).

Tier	Result and evidence	Run or source
B1 AI-mode	PASS	AI-mode run `387591f0`
B2 MCP	PASS	`data.releases.v1` over MCP, same run
B3 RAG	PASS	Run `56c85266`, 2 citations
B4 Multi-Agent	PASS	Run `2058658b`, rejected in the UI
B5 Endpoint security	58 PASS, 0 FAIL	Cloud Deployment run 40041985429
B6 Data security	43 PASS, 0 FAIL	Laptop, same deployment, 24 October

B5 checks TLS 1.2+, the HTTPS redirect, HSTS, CSP and framing headers, the site login, and operator credentials on AI and workflow routes. B6 checks Key Vault RBAC and purge protection, least-privilege managed identity, encrypted disks, OIDC without secrets, and internal database networks.

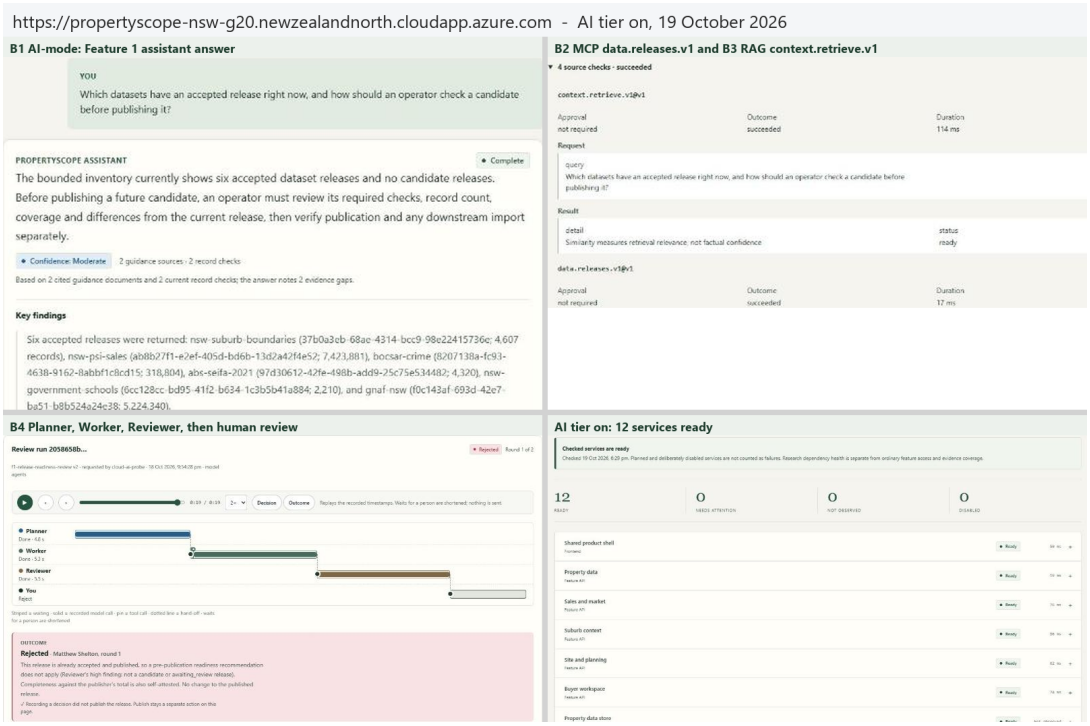


Figure 11 B1–B4 through the Azure UI

14 Showcase video

Showcase video: a pre-recorded walkthrough in which all five members, Matthew Shelton, Burhan Naeem, James Huang, Michael White and Derek Song, present their work.